FORMULA DI STIRLING

$$\lim_{n\to\infty} \frac{n!}{n^n e^{-n}\sqrt{2\pi n}} = 1$$

ESEMPIO

$$\lim_{n\to+\infty} \frac{n!\cdot 3^n}{n^n}$$

$$\frac{n!\, e^{-n}\sqrt{2\pi n}\cdot 3^n}{n^n \cdot e^{-n}\sqrt{2\pi n}} = 1\cdot e^{-n}\sqrt{2\pi n}\cdot 3^n$$

($n^n e^{-n}\sqrt{2\pi n} \to 1$)

$$\left(\frac{3}{e}\right)^n \cdot \sqrt{2\pi n} \to +\infty$$

↳ $\frac{3}{e} > 1 \to +\infty$

$$1\cdot(+\infty)(+\infty) = +\infty$$

COMPLESSITÀ ASINTOTICA

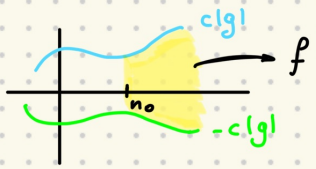
NOTAZIONE

O - GRANDE

Confronto differenti funzioni

$f, g: \mathbb{N} \to \mathbb{R}$

$f \in O(g)$ se $\exists\, n_0 \in \mathbb{N},\ c>0 :\ |f(n)| \le c\cdot|g(n)| \quad \forall n \ge n_0$



ESEMPIO

I. $O(1) \ni f$, f è limitata

$\exists c>0,\ n_0 \in \mathbb{N}$

$0 \le |f(n)| \le c\cdot 1 = c \quad \forall n \ge n_0$

II. $f(n) = 3n^2+10n \in O(n^2)$

$f(n) = 3n^2+10n \le 3n^2+10n^2$

$\le 13n^2 \quad \forall n \in \mathbb{N}$

$c = 13,\ n_0 = 0$

RELAZIONE INVERSA

NOTAZIONE

Ω - GRANDE

$f, g: \mathbb{N} \to \mathbb{R}$

$f \in \Omega(g)$ se $\exists n_0 \in \mathbb{N},\ c>0 :\ 0 \le c|g(n)| \le |f(n)| \quad \forall n \ge n_0$

ESEMPIO

I. $f(n) = 3n^2+10n \in \Omega(n^2)$

$f(n) = 3n^2+10n \ge 3n^2 \quad \forall n \in \mathbb{N}$

$c = 3,\ n_0 = 0$

11. $n^2 \notin \Omega(n^3)$

se $n^2 \in \Omega(n^3)$ allora $\exists\, c>0$ $n_0 \in \mathbb{N}$

$0 \le c \cdot n^3 \le n^2 \quad \forall n \ge n_0 > 0$

$\frac{0}{n^2} \le c \cdot \frac{n^3}{n^2} \le \frac{n^2}{n^2}$

$0 \le c \cdot n \le 1 \quad \forall n \ge n_0$

↑ DIVERGE E NON È LIMITATA

QUINDI PER ASSURDO

$n^2 \notin \Omega(n^3)$

COMBINAZIONE TRA Ω e O

NOTAZIONE Θ - GRANDE

$f, g: \mathbb{N} \to \mathbb{R}$ $\quad f \in \Theta(g)$ se $\exists\, n_0 \in \mathbb{N},\ c_1>0, c_2>0$

$0 \le c_1 |g(n)| \le |f(n)| \le c_2 |g(n)| \quad \forall n \ge n_0$

$\Downarrow$

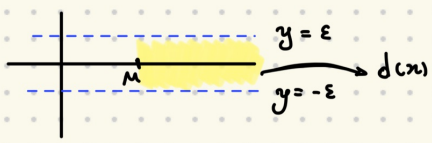
$f \in O(g)$

$f \in \Omega(g)$

FUNZIONI E LIMITI

$d: \mathbb{R} \to \mathbb{R}$ si dice infinitesima per $x \to +\infty$

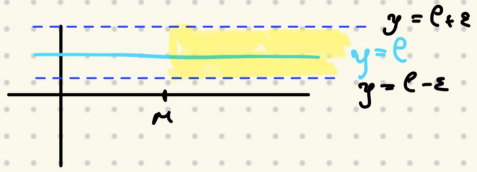
$\forall \varepsilon > 0\ \exists M \in \mathbb{R}: \quad 0 \le |d(x)| < \varepsilon \quad \forall x > M$



LIMITE

$f: \mathbb{R} \to \mathbb{R}$ ha limite $\ell \in \mathbb{R}$ per $x \to +\infty$ se $f(x) - \ell$ è infinitesima per $x \to \infty$

$\forall \varepsilon > 0\ \exists M \in \mathbb{R}: |f(x) - \ell| < \varepsilon \quad \forall x > M$



$\lim_{x \to \infty} \frac{x^2}{x^2+1} = 1$

$\frac{x^2(1)}{x^2(1+\frac{1}{x^2})} = 1$

$\forall \varepsilon > 0 \quad \left| \frac{x^2}{x^2+1} - 1 \right| < \varepsilon$

$= \left| \frac{x^2 - x^2 - 1}{x^2+1} \right|$

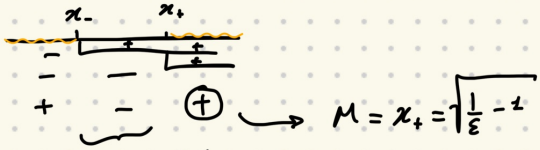
$= \left| \frac{-1}{x^2+1} \right|$

$= \frac{1}{x^2+1} < \varepsilon \iff \frac{1}{\varepsilon} < x^2+1$

$\frac{1}{\varepsilon} - 1 < x^2$

$x^2 = \frac{1}{\varepsilon} - 1$

$x = \pm\sqrt{\frac{1}{\varepsilon} - 1} = x_\pm$



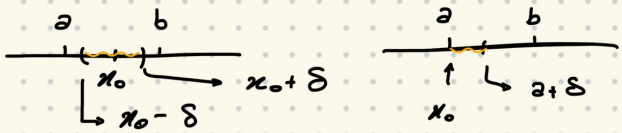
$\to M = x_+ = \sqrt{\frac{1}{\varepsilon} - 1}$

Non è valida la disuguaglianza

DEF

$d: (a,b) \to \mathbb{R}, \quad x_0 \in [a,b]$

d è infinitesima per $x \to x_0$ se $\forall \varepsilon > 0 \ \exists \delta > 0: |d(x)| < \varepsilon \quad \forall \ 0 < |x - x_0| < \delta$



LIMITE DI UNA FUNZIONE

$f: (a,b) \to \mathbb{R}, \quad x_0 \in [a,b]$

diciamo che $f(x) \to \ell \in \mathbb{R} \ \forall x \to x_0$ se
$f(x) - \ell$ è infinitesima $\forall x \to x_0$

$\forall \varepsilon > 0 \ \exists \delta > 0: |f(x) - \ell| < \varepsilon \quad \forall \ 0 < |x - x_0| < \delta$

ESEMPI

1. $\lim_{x \to 1} \overbrace{3x+5}^{f(x)} = 8 \to \ell$

$3(1) + 5 = 8$

$|f(x) - \ell| = |3x + 5 - 8| = |3x - 3|$

$= |3(x-1)|$

$|f(x) - \ell| = \underbrace{3 \cdot |x-1|}_{< 3\delta} < \varepsilon$

$\forall |x-1| < \delta$

$\delta = \frac{\varepsilon}{3}$ allora $|f(x) - 8| < \varepsilon \cdot \delta = \not{3} \frac{\varepsilon}{\not{3}} = \varepsilon$

II. $\lim_{x \to x_0} \sin(x) = \sin(x_0)$

$\sin(x_0)$

$x \in [0, \frac{\pi}{2}]$

$|\sin(x) - \sin(x_0)| = |2 \sin(\frac{x-x_0}{2}) \cos(\frac{x-x_0}{2})|$ → Formula di prostaferesi

$\leq 2 |\sin(\frac{x-x_0}{2})| \leq 2 |\frac{x-x_0}{2}| = |x-x_0|$

↑ $|\cos(y)| \leq 1$ ↑ $|\sin(y)| \leq y$

$\delta = \varepsilon$

III. $\lim_{x \to x_0} \cos(x) = \cos(x_0)$

$\cos(x_0)$

$x \in [0, \frac{\pi}{2}]$

$|\cos(x) - \cos(x_0)| = |-2 \sin(\frac{x-x_0}{2}) \sin(\frac{x+x_0}{2})|$

$\leq 2 |\sin(\frac{x-x_0}{2})| \leq 2 |\frac{x-x_0}{2}| \leq |x-x_0|$

↑ $|\sin(\frac{x+x_0}{2})| \leq 1$ ↑ $|\sin(y)| \leq |y|$

$\delta = \varepsilon$

PROPRIETÀ DEI LIMITI

f, g

$\lim_{x \to x_0} f(x) = \ell$

$\lim_{x \to x_0} g(x) = m$

$\Rightarrow$

$\lim_{x \to x_0} f(x) + g(x) = \ell + m$

$\lim_{x \to x_0} f(x) \cdot g(x) = \ell \cdot m$

$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{\ell}{m} \quad m \neq 0$

$f(x) \leq g(x) \Rightarrow \lim_{x \to x_0} f(x) \leq \lim_{x \to x_0} g(x) \quad \forall x$

$\ell \leq m$

DIM

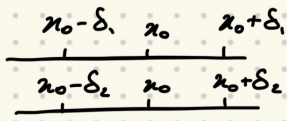
$\forall \varepsilon > 0 \ \exists \delta_1 > 0 : |f(x) - \ell| < \varepsilon \quad \forall 0 < |x - x_0| < \delta_1$

$\forall \varepsilon > 0 \ \exists \delta_2 > 0 : |g(x) - m| < \varepsilon \quad \forall 0 < |x - x_0| < \delta_2$

↓ quindi anche $\forall \frac{\varepsilon}{2} > 0$ — Anche $\frac{\varepsilon}{2}$

$\forall \varepsilon > 0 \ \exists \delta > 0 : |f(x) + g(x) - (\ell + m)| < \varepsilon \quad \forall 0 < |x - x_0| < \delta$

Prendiamo $\delta = \min(\delta_1, \delta_2)$



$|f(x) + g(x) - (\ell + m)| = |f(x) - \ell + g(x) - m|$

$= |(f(x) - \ell) + (g(x) - m)| \leq |f(x) - \ell| + |g(x) - m| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$

→ $|a+b| \leq |a| + |b|$

ESEMPIO

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1$$ ← LIMITE NOTEVOLE

$$\frac{\sin(x)}{x} = 1$$

$$\rho \quad \frac{\sin(0)}{0} = \frac{0}{0}$$

⇓

$x \in [0, \frac{\pi}{2}]$

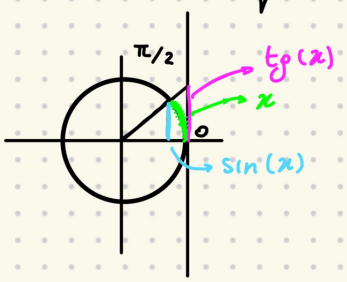
Pari?

$f(-x) = f(x)$

$$\frac{\sin(-x)}{-x} = \frac{\sin(x)}{x}$$

$$-\frac{\sin(x)}{-x} = \frac{\sin(x)}{x}$$

$$\frac{\sin(x)}{x} = \frac{\sin(x)}{x}$$ ✓



$$\frac{\sin(x)}{\sin(x)} \le \frac{x}{\sin(x)} \le \frac{tg(x)}{\sin(x)}$$

$x \in (0, \frac{\pi}{2})$

$$1 \le \frac{x}{\sin(x)} \le \frac{\sin(x)}{\cos(x)} \cdot \frac{1}{\sin(x)}$$

$\frac{1}{\cos(x)}$

$$1 \le \frac{x}{\sin(x)} \le \frac{1}{\cos(x)}$$

$$\lim_{x\to 0} 1 = 1$$

$$\lim_{x\to 0} \frac{1}{\cos(x)} = 1$$

$$1 \le \frac{x}{\sin(x)} \le 1$$ TEOREMA DEI CARABINIERI

⇓

$$\lim_{x\to 0} \frac{x}{\sin(x)} = 1$$

CONSEGUENZE

- $$\lim_{x\to 0} \frac{tg(x)}{x} = 1$$

$$\rho \quad \frac{\sin(x)}{x} \cdot \frac{1}{\cos(x)}$$

↳ 1

$$\rho \quad \frac{1}{\cos(x)} = \frac{1}{1} = 1$$

- $$\lim_{x\to 0} \frac{1-\cos(x)}{x} = 0$$

$$\frac{1-1}{0} = \frac{0}{0}$$ INDETERMINATA

$$\rho \quad \frac{1-\cos(x)}{x} \cdot \frac{1+\cos(x)}{1+\cos(x)}$$

$$\rho \quad \frac{1+\cos^2(x)}{x(1+\cos(x)}$$

- $\lim_{x\to 0} \frac{1-\cos(x)}{x^2} = \frac{1}{2}$

$\frac{0}{0}$ INDETERMINATA

ρ $\frac{1-\cos(x)}{x} \cdot \frac{1+\cos(x)}{1+\cos(x)}$

ρ $\frac{1-\cos^2(x)}{x^2(1+\cos(x))}$

ρ $\frac{1-\sin^2(x)}{x^2(1+\cos(x))}$

ρ $\frac{\sin^2 x}{x^2} \cdot \frac{1}{1+\cos(x)}$

ρ $\frac{\sin(x)}{x} \to 1 \cdot \frac{\sin(x)}{x} \to 1 \cdot \frac{1}{1+\cos(x)}$

ρ $1 \cdot 1 \cdot \frac{1}{2} = \frac{1}{2}$

ρ $\frac{\sin^2(x)}{x} \cdot \frac{1}{(1+\cos(x))}$

ρ $\frac{\sin(x)}{x} \to 1 \cdot \frac{\sin(x)}{1+\cos(x)}$

ρ $1 \cdot \frac{0}{2} = 0$

LIMITI NOTEVOLI

- $\lim_{x\to 0} \frac{\sin(x)}{x} = 1$
- $\lim_{x\to 0} \frac{tg(x)}{x} = 1$
- $\lim_{x\to 0} \frac{1-\cos(x)}{x} = 0$
- $\lim_{x\to 0} \frac{1-\cos(x)}{x^2} = \frac{1}{2}$
- $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x\to 0} x^\alpha \ln(x) = 0 \quad \forall \alpha > 0$
- $\lim_{x\to \infty} \frac{x^\alpha}{a^x} = 0 \quad \forall a > 1, \alpha > 0$